Training and Testing Role Transfer in Language Models

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1 Purpose and Aims

What if a model has seen Marco and Polo interact with different people? Can it tell who acted on whom when they first appear together? A model may recognize both names and the action while reversing their roles. I want to understand how Transfomer models represent roles such as who perform an action and who receives it, and when these representations help them interpret new combinations. A wrong role assignment changes the meaning of a sentence, even if the model performs well on familiar tests. My research questions are the following:

1. **RQ1:** How does the pattern of observed relations during training affect whether a small Transformer transfers roles to character pairs it has not seen together?
2. **RQ2:** What patterns of activity inside the model distinguish succesful from failed role transfer, and can a targeted change in training improve this transfer
3. **RQ3:** How consistent are pretrained language models' role answer across different sentence forms and question wordings, and do their internal responses remain consistent as well?

I will begin with a controlled tasj, where I know which relations the model has seen during training, I will then test whether similar patterns appear in pretrained language models, using constructed sentence pairs and naturally occurring sentence pairs from a annotated dataset. I will also examine the models' internal responses where this is possible. The controlled study is the core of the project, while the pretrained language model experiments test how far its findings extend to more realistic language settings. Together, these experiments aim to show what a correct answer does, and does not, reveal about how a model represents linguistic roles.

2 Survey of the Field

Models can understand familiar words but struggle when those words appear in new combinations. Kim and Linzen (2020) showed this problem in COGS, where models performed much better on ordinary test examples than on systematic new combinations. Zhou et al. (2023) showed that the diversity and repetition of training examples affect "compositional generalization", which is the ability to handle new combinations of familiar parts. Patil et al. (2024) also showed that language models can sometimes generalize from indirect evidence when a linguistic construction is absent from training. From these studies I take the need to look at both unseen combinations and the evidence available during learning. This raises another question: if the amount of training stays fixed, does the pattern of observed relations affect transfer to a pair that was never seen together? My expectation is that it could, because different arrangements can give the model different indirect evidence about how the same participants behave across relations, even when the total amount of training is unchanged.

Such a test is only useful if the training examples support a clear answer, Jabbar et al. (2025) showed that some apparent failures can arise from ambiguous or contradictory learning targets. For that reason I will check whether the missing pair task, where two familiar characters are tested together for the first time, has a clear answer before comparing models. This also motivates separating a failure to transfer roles from a task where the required answer could not be learned from the available evidence.

The internal analysis builds on work about how models represent roles, Denning et al. (2026) found role information in some attention heads, which are components that control how information from different parts of an input is combined, while Aljaafari et al. (2026) studied groups of components involved in semantic roles. These findings motivate examining internal activity and not only the models' answers. Finding role information somewhere inside a model, however, would not by itself show whether that information supports transfer. My study asks if internal activity differs when role transfer succeeds or fails, and whether changing a relevant internal response can affect the models' answer.

Evaluation of pretrained models also needs care, Zhuo et al. (2024) studied sensitivity to prompt wording, while Deng et al. (2024) investigated overlap between evaluation data and model training data. I will compare similar sentence forms and question wordings in pretrained models, using newly constructed items and selected natural sentences. These tests can show if an observed pattern extends beyond the controller task, although they cannot establish what the pretrained models saw during training.

3 Project Description

3.1 Theories and methods

The experiments below build on the approaches described in the previous section, adapting them to test the three research questions of this project.

Controlled task (RQ1). I will begin with a small, two layer Transformer because I can control what it learns and examine its internal activity. A Transformer uses attention to combine information from different parts of its input. The model receives a controlled message, such as "the fox moves towards the dog," and chooses among four scenes: the fox approaches the dog, the dog approaches the fox and both directions of moving away, these choices let me separate mistakes about the action from mistakes about who moves, while knowing which relations appeared during training arrangement. The controlled space has ten characters, giving 45 possible unordered pairs. The model will train on twenty pairs and tested on the other twenty five. It may see the fox with the cat and the dog with the bird, but never the fox with the dog. The characters and actions are familiar, but their combination is new. I will test every missing pair in both directions and change which character appears first, so that neither one pair nor a fixed position can determine the answer. The training pairs form a network, where a connection means that two characters appeared together. I will change these connections while keeping the number of examples and the frequency of characters and actions fixed. The idea is to test whether different arrangements of the same amount of evidence affect transfer. I plan four layouts with three matched starts for each comparison, followed by new layouts to check whether a finding repeats. I will also check that the questions have clear answers and that other factors do not affect the final result. I will report performance on familiar and missing pairs, including action errors and role reversals. Familiar pair performance shows if the model learned the task, while missing pair performance shows if it transfers that learning, A difference that repeats on new layouts would support an effect of the training.

Internal activity and training comparison (RQ2). I want to examine if the model's internal activity helps explain why it transfers some missing pairs correctly but reverses the roles in others. I will check whether this activity distinguishes who moves in familiar pairs, and then compare the same patterns between successful and failed missing pairs. If I find a clear difference I will change the relevant internal response using a matched example and test if the mode's answer changes. If the first check fails, I will not treat it as evidence that the model has no role information. I will also test a possible way to improve transfer. Both groups of models will receive the same examples, including both directions of each observed pair, one group will learn with the usual four choice task and the other will have an additional rule encouraging consistent answers, when characters exchange roles. This changes the training constraints. Instead of adding examples, I will keep the model and training schedule the same and choose the strength of the rule before the final test. Then I will compare both groups on new layouts, using the same initial

settings for each pair. I will report missing pairs, familiar pairs, and action accuracy. If the improvement holds across layouts, it would help support the training rule.

Pretrained language-model experiment (RQ3). I will examine Qwen2.5-1.5B-Instruct directly and compare its answers with two larger models available through Berget AI: Gemma 4 31B and Qwen3.8 27B (Qwen Team, 2024; Google DeepMind, 2026; Qwen Team, 2026). I can inspect internal activity in the smaller model, while the larger models let me test whether its answer patterns also appear in more capable systems. I will confirm that the hosted models are still available before the experiment. I will use about 120 development events to fix the prompts and scoring rules, followed by 400 new test events in four sets of 100. Each event will appear in active and passive form and with two question wordings. For example, "Marco chased Polo" and "Polo was chased by Marco" should both lead to "Marco." I will balance names, verbs, and actor position so that a simple positional shortcut cannot solve the task. Every model will receive the same items with fixed settings and scoring rules. I will report accuracy for each sentence form, agreement between forms, and changes caused by question wording. Agreement will be measured separately from accuracy, since two answers can agree and still be wrong. A difference that repeats across sets and models would provide stronger evidence than one appearing in only one set or model. I will also select about 200 naturally occurring sentences from the QA-SRL gold evaluation data (Roit et al., 2020). keeping cases with two clear participants and checking their role answers myself. I will report these separately because natural sentences contain structures and vocabulary absent from the constructed examples. On a smaller active-passive set, I will also examine whether Qwen's internal responses track the actor across sentence forms. Since the pretrained models' exposure to similar sentences is unknown, these tests concern the reliability of their answers rather than proving that every example is new to them.

Evaluation. Before testing, I will check the role labels in the constructed and QA-SRL items and remove ambiguous cases using written rules. The main measure will be exact role-pair accuracy, supported by individual role accuracy, role reversals, unclear answers, and consistency across sentence forms and question wordings. I will compare versions of the same event as paired observations and report accuracy differences with 95% confidence intervals over events, keeping all versions of one event together. Planned active-passive comparisons will use adjusted p-values. For the controlled task, I will report results by network layout and random start rather than treating generated scenes as independent runs.

3.2 Time plan and implementation

The project starts in October and finishes in mid-January. Here I draft the methods and results sections as the experiments finish, so that completed text accumulates throughout the timeline.

1. **October (after 7th Oct deadline):** I will do the controlled task, check that the missing pair answers are identifiable and decide the measures before running experiments.
2. **From late October to mid-November:** I will run the main transfer comparison across new layouts and random starts, do a draft of the RQ1 results as the runs finish
3. **Mid to late November:** examine internal activity and test the training change on new layouts, draft the RQ2 results.
4. **Late November to early December:** I will run the pretrained model experiments on constructed and selected natural sentences, and draft the RQ3 results.
5. **By December 14th:** Finish introduction, discussion and the remaining sections to produce a first draft paper ready for peer review
6. **December to 15th January:** Use peer feedback to revise the paper and prepare for the final presentation

If the controlled experiments take longer than expected, I will reduce the pretrained-model extensions first. The controlled transfer comparison and RQ2 analysis remain the priority, followed by the core pretrained-model comparison needed for RQ3.

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